Benjamí Moreno Torres

Data Science & AI Lead
Berlin, Germany · +49 172 671 9415 · benji.moreno@gmail.com
linkedin.com/in/benjami-moreno-torres · benjami-moreno-torres.com



Summary

Data and AI leader with 10+ years applying machine learning to R&D problems in materials science, structural monitoring, and market intelligence. Trained as a mathematician and an architect, I specialise in sequential learning for materials discovery, ontology engineering, and NLP, backed by hands-on Python and MLOps practice. At TU Berlin I led the building-demonstrations work package of a Horizon Europe project across six partners in four countries, alongside the project's AI and data research. I am looking for a Data Science or AI Lead role in an organization with strong R&D.

Experience

Research Associate, AI & Data · Work Package Leader, Technische Universität Berlin
2023/04 – 2026/05

Berlin, Germany · *Horizon Europe project REINCARNATE (circular construction materials)*

- Led the *Demonstrations in Buildings* work package, coordinating 6 partner organizations across 4 countries and owning deliverables, timelines, and integration of results.
- Co-authored **SLAMD** (Sequential Learning App for Materials Discovery), an open-source active-learning tool for sustainable materials design (NanoWorld Journal, 2023), and coordinated its industrial application in the project.
- Designed an ontology-based data architecture (OWL/RDF/SPARQL) for FAIR interoperability across heterogeneous research datasets.
- Developed probabilistic life-cycle models for construction products; first author of project deliverable D1.3 on probabilistic methods for life-cycle assessment.
- Supervised 4 graduate researchers in AI and digital construction, guiding projects from scoping to publication-ready outputs.
- Managed stakeholder communication across academia, industry, and EU project officers.

Machine Learning Researcher, BAM – Federal Institute for Materials Research and Testing
2020 – 2023/01

Berlin, Germany

- Built predictive ML models (clustering, regression, ensembles) for material durability and circularity assessment at one of Europe's leading materials research institutes.
- Structured and analyzed databases on Alkali-Activated Concretes from heterogeneous experimental sources, enabling reproducible ML workflows.
- Developed data pipelines for EU digitalization projects in materials science, improving cross-institutional data quality and accessibility.
- Collaborated with interdisciplinary teams of chemists, physicists, and engineers to translate domain expertise into ML feature engineering.

Data Scientist & ML Engineer, 3S'TECH S.L.
2016 – 2020

Girona, Spain · *Structural Health Monitoring (SHM)*

- Developed predictive algorithms for real-time structural health monitoring using IoT sensor networks on civil infrastructure.
- Designed end-to-end data processing pipelines for sensor data ingestion, cleaning, transformation, and anomaly detection.
- Implemented ML models for engineering databases and concrete testing, improving assessment accuracy for structural integrity.

Data Scientist — NLP & Analytics, Symanto GmbH
2012/08 – 2016/05

Nürnberg, Germany

- Built Python and C# tools for social media sentiment analysis and market research, processing large-scale unstructured text data.
- Contributed to NLP pipelines (text classification, entity extraction) and graph-based influencer analytics for enterprise clients.
- Delivered actionable data insights for digital marketing campaigns, translating technical analysis into business recommendations.

Structural Designer & Architectural Engineer, Various firms 2005 – 2011
Spain

- Structural design, planning, and valuation across residential, heritage, and public projects including the Disseny Hub Barcelona.
- Developed the systems thinking, spatial reasoning, and multi-constraint optimization skills that now shape my approach to data architecture and ML system design.

Selected Project

SLAMD — Sequential Learning App for Materials Discovery
BAM / TU Berlin · open source · Role: Co-developer, tester and co-author · Field: Materials science

- Active-learning tool for the inverse design of sustainable cementitious materials; it selects the most informative experiments, reducing the lab tests needed to find optimal mixes.
- Contributed to design, validation and the 2023 NanoWorld Journal publication; code at github.com/BAMresearch/WEBSLAMD, live demo at slamd-demo.herokuapp.com.

Technical Skills

ML / AI: Sequential / Active Learning, NLP, Graph Analytics, Clustering, Regression, Ensemble Methods, LSTMs, Probabilistic Models (Markov Chains, Bayesian Networks)
MLOps & Production: Docker, Kubernetes, MLflow, DVC, Airflow, FastAPI, BentoML, Prometheus, Grafana, CI/CD, Git/GitHub, LangChain
Data & Semantic Web: Ontologies (OWL, RDF, SPARQL), FAIR Data, Linked Data, Data Pipeline Design
Programming: Python (primary), R, C#, SQL
Leadership: Horizon Europe, International consortium coordination, Multi-stakeholder management, Graduate supervision, Technology transfer

Certifications

MLOps Specialty (Machine Learning Engineer) (Liora / DataScientest, certified by Université Paris La Sorbonne) 2026/10

Education

PhD, Semantic Web Applications for Materials Discovery Ongoing
Technische Universität Berlin

Master in Architecture (375 ECTS) 2011
Universitat de Girona

Master in Mathematics for Industry (60 ECTS) 2001
Universitat Autònoma de Barcelona

Degree in Mathematics (300 ECTS) 2000
Universitat Autònoma de Barcelona

Selected Publications

An Ontology-Based Approach to Enable Data-Driven Research in the Field of NDT in Civil Engineering
Moreno Torres B., Völker C., Nagel S. M., Hanke T., Kruschwitz S. · *Remote Sensing 13(12), 2426* · 2021

Concreting a sustainable future: A dataset of alkali-activated concrete and its properties
Moreno Torres B., et al. · *Data in Brief* · 2023

Presenting SLAMD — A Sequential Learning Based Software for the Inverse Design of Sustainable Cementitious Materials
Völker C., Moreno Torres B., et al. · *NanoWorld Journal* · 2023

Data Driven Design of Alkali-Activated Concrete using Sequential Learning
Völker C., Moreno Torres B., et al. · *Journal of Cleaner Production 418, 138221* · 2023

Languages

Spanish & Catalan: Native
English: Professional proficiency
German: B2 (strong passive skills, actively improving)